

Write Algebraic Expressions

Lesson 6-3

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



x

stands for a number

In $5t + 20$, t could be hours worked — if $t = 3$, the expression equals 35; if $t = 8$, it equals 60

Variable

Write the definition:

$3x + 5$

no equal sign

$3n + 7$ means 'triple a number, then add 7' — it has a variable (n), a coefficient (3), and a constant (7)

Algebraic Expression

Write the definition:



4x

coefficient = 4

In $8x$, the coefficient 8 means '8 groups of x ' — if $x = 3$, then $8x = 8 \times 3 = 24$

Coefficient

Write the definition:



$2x + 5$

constant = 5 (fixed)

In $3n + 7$, the 7 never changes no matter what n is — like a flat fee that stays the same

Constant

Write the definition:

$$3x + 2x = 5x$$

same variable → combine

$4x + 2x = 6x$ (like terms, same variable); but $4x + 2y$ cannot be combined (different variables)

Like terms

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can write algebraic expressions from words and real-world situations.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Variable

Algebraic Expression

Coefficient

Constant

Like terms

1

A letter that stands for an unknown number is a _____.

2

A math phrase with variables, numbers, and operations but no equal sign is an _____.

3

The number multiplied by a variable, like the 7 in $7n$, is the _____.

4

A term that is just a number and never changes, like the 5 in $3x + 5$, is a _____.

5

Terms with the same variable part are _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Which expression represents 'the product of 6 and a number n '?

- A. $6n$
- B. $6 + n$
- C. $n - 6$
- D. $n \div 6$

Step 1 'Product' means multiplication, so the product of 6 and n is $6n$.

 **Answer:** A. $6n$

 We do — together

Solve this one with your class using the same steps.

Problem: Which expression represents '9 less than a number y '?

- A. $y - 9$
- B. $9 - y$
- C. $y + 9$
- D. $9y$

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Which expression represents 'the sum of a number m and 15'?

- A. $m + 15$
- B. $m - 15$
- C. $15m$
- D. $m \div 15$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Line 2: The chorus says '5 less than a number of tracks x .' Which expression matches?

A. $x - 5$

B. $5 - x$

C. $5x$

D. $x + 5$

Show your work:

2. Line 3: A verse says 'twice a number of speakers' (use s). Which expression matches?

A. $2s$

B. $s + 2$

C. $s - 2$

D. $s \div 2$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Write a real-world situation that could be modeled by the expression $4n + 10$. Explain what the variable n represents, what the coefficient 4 means, and what the constant 10 represents.

Sentence starter: Situation: _____. The variable n represents _____. The coefficient 4 means _____ for each _____. The constant 10 represents _____.

Show your work:

Reflect — Exit Ticket

Which expression represents '3 more than twice a number n '?

- A. $2n + 3$
- B. $3n + 2$
- C. $2(n + 3)$
- D. $2 + 3n$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Variable
2. **Notes 2:** Algebraic Expression
3. **Notes 3:** Coefficient
4. **Notes 4:** Constant
5. **Notes 5:** Like terms
6. **We Try (notes):** A. $y - 9$ — *'Less than' means subtract from the number, so it's $y - 9$.*
7. **You Try (notes):** A. $m + 15$ — *'Sum' means addition, so the sum of m and 15 is $m + 15$.*
8. **Try It 1:** A. $x - 5$ — *'Less than' subtracts from the number: start with x and take away 5 , so $x - 5$.*
9. **Try It 2:** A. $2s$ — *'Twice' means two times the number, so twice s is $2s$.*
10. **Exit Ticket:** A. $2n + 3$ — *'Twice a number n ' is $2n$. '3 more than' means $+ 3$. So the expression is $2n + 3$.*